A Single Stepsize Suffices for Unprojected Linear TD(0)

Simultaneous Robust and Fast Rates via Polyak–Ruppert Averaging

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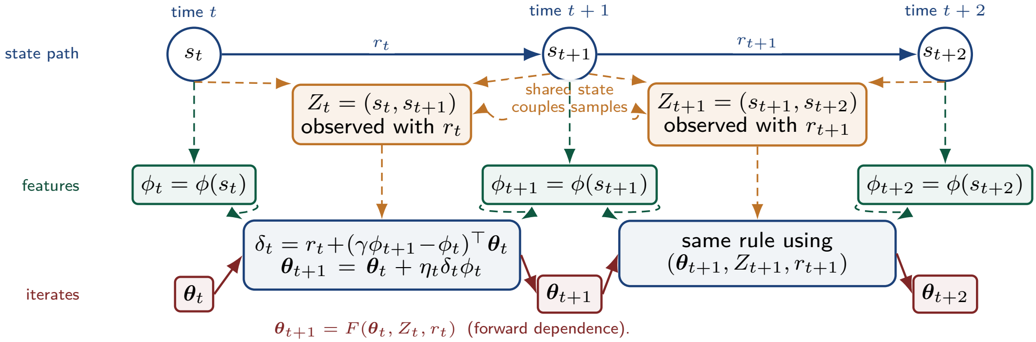
Setting: Linear TD(0) with Markovian sampling

A fixed policy gives one **dependent** trajectory. Linear TD(0) combines the current iterate with the current transition sample:

$$
\begin{aligned}
Z_t &= (s_t, s_{t+1}), \\
V_{\boldsymbol{\theta}}(s) &= \phi(s)^\top \boldsymbol{\theta}, \qquad \|\phi(s)\| \le \phi_\infty, \\
\delta_t(\boldsymbol{\theta}_t) &= r_t + \gamma \phi(s_{t+1})^\top \boldsymbol{\theta}_t - \phi(s_t)^\top \boldsymbol{\theta}_t, \qquad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \eta_t \delta_t(\boldsymbol{\theta}_t)\phi(s_t).
\end{aligned}
$$



Performance measure: TD potential and two regimes

For $v \in \mathbb{R}^{|\mathcal{S}|}$, define the stationary value norm and Dirichlet seminorm

$$\|v\|_{\boldsymbol{D}}^2 := \sum_s \pi(s) v(s)^2, \qquad \|v\|_{\mathrm{Dir}}^2 := \frac{1}{2} \sum_{s,s'} \pi(s) P^{\mu}(s,s') \big(v(s) - v(s')\big)^2.$$

We measure error by the TD potential

$$f(\boldsymbol{\theta}) = (1-\gamma) \left\| \boldsymbol{V_\theta} - \boldsymbol{V_{\theta^*}} \right\|_{\boldsymbol{D}}^2 + \gamma \left\| \boldsymbol{V_\theta} - \boldsymbol{V_{\theta^*}} \right\|_{\mathrm{Dir}}^2.$$

It has curvature around the TD fixed point $\boldsymbol{\theta}^*$:

$$f(\boldsymbol{\theta}) - f(\boldsymbol{\theta}^*) \geq \omega \left\| \boldsymbol{\theta} - \boldsymbol{\theta}^* \right\|^2, \qquad \omega := (1-\gamma) \lambda_{\min}(\boldsymbol{\Phi}^\top \boldsymbol{D} \boldsymbol{\Phi}) > 0.$$

Two finite-time regimes

Robust rate	**Fast rate**
curvature-free	curvature-dependent
$\widetilde{\mathcal{O}}(1/\sqrt{T})$	$\widetilde{\mathcal{O}}(1/(\omega T))$

Why stability is hard under Markovian sampling

Two coupled difficulties.

- **Not i.i.d.:** $Z_t = (s_t, s_{t+1})$ is generated by the same trajectory that generated $\boldsymbol{\theta}_t$. Thus, in general,
$$\mathbb{E}[\boldsymbol{g}(\boldsymbol{\theta}_t, Z_t) \mid \mathcal{F}_{t-1}] \neq \bar{\boldsymbol{g}}(\boldsymbol{\theta}_t).$$
- **Self-amplification:**
$$\|\boldsymbol{g}(\boldsymbol{\theta}_t, Z_t)\| \leq r_\infty \phi_\infty + 2\phi_\infty^2 \|\boldsymbol{\theta}_t\|.$$
A loose bound on $\|\boldsymbol{\theta}_t\|$ makes the update bound loose, which feeds back into $\boldsymbol{\theta}_{t+1}$.

What prior work uses

- **Projection:** (Bhandari et al. 2018; Liu and Olshevsky 2021). Stable by design, but changes the TD algorithm and requires prior knowledge of a suitable projection radius.
- **Curvature / contractive stability:** (Srikant and Ying 2019; Patil et al. 2023; Samsonov et al. 2024; Mitra 2025; Chandak and Borkar 2025; Durmus et al. 2025; Li et al. 2026). Projection-free or lightly modified, but the rate, burn-in, or tuning can depend on the curvature.

Question: one plain TD stepsize for both regimes?

Question

Can unprojected TD(0) achieve high-probability stability and simultaneous robust and fast rates without knowledge of ω?

Best of both worlds target

$$f(\bar{\boldsymbol{\theta}}_T) - f(\boldsymbol{\theta}^*) \le \widetilde{\mathcal{O}}\left(\min\left\{\frac{1}{\sqrt{T}}, \frac{1}{\omega T}\right\}\right)$$

- If ω is small: still keep a $1/\sqrt{T}$ guarantee.
- If ω is large: automatically obtain the fast $1/(\omega T)$ rate.

Main results: stability and simultaneous rates

Use a single decaying stepsize

$$\eta_t = \frac{1}{c\,\tau_{\mathrm{mix}}\,\phi_\infty^2}\,\frac{1}{\sqrt{t+1}\,\log(t+3)}.$$

Let

$$R_{\mathrm{base}} = \max\left\{\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|\,,\|\boldsymbol{\theta}^*\|\,,\frac{r_\infty}{\phi_\infty}\right\}.$$

Define the weighted Polyak–Ruppert average

$$S_T = \sum_{t=1}^{T} \eta_{t-1}, \qquad \bar{\boldsymbol{\theta}}_T = \frac{1}{S_T}\sum_{t=1}^{T} \eta_{t-1}\boldsymbol{\theta}_{t-1}.$$

Mixing assumption

Assume the transition chain $Z_t = (s_t, s_{t+1})$ is irreducible and aperiodic with stationary law π_Z. Then,

$$\max_{z\in\mathcal{Z}} \left\| P_Z^k(z,\cdot) - \pi_Z \right\|_{\mathrm{TV}} \le 4\cdot 2^{-k/\tau_{\mathrm{mix}}}, \qquad k \ge 0.$$

Main theorem

For any sufficiently large c depending on δ, set $R_{\max} = \rho(c,\delta)R_{\mathrm{base}}$. With probability at least $1-\delta$, for all $T \ge 1$,

$$\sup_{t\ge 0}\|\boldsymbol{\theta}_t\| \le R_{\max}, \qquad f(\bar{\boldsymbol{\theta}}_T) - f(\boldsymbol{\theta}^*) \le \widetilde{\mathcal{O}}\left(R_{\max}^2 \min\left\{\frac{\tau_{\mathrm{mix}}^2\phi_\infty^4}{\omega T}, \frac{\tau_{\mathrm{mix}}\phi_\infty^2}{\sqrt{T}}\right\}\right).$$

Stability proof I: make the loop self-bounding

Use induction on the past event

$$\mathsf{IH}_{t-1} := \left\{ \max_{0 \le i < t} \|\boldsymbol{\theta}_i\| \le R_{\max} \right\}.$$

Since $\|\boldsymbol{\theta}_t\| \le \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\| + \|\boldsymbol{\theta}^*\|$, we can use $\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|$ to control $\|\boldsymbol{\theta}_t\|$. Expanding $\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2$ gives

$$\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2 = \|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|^2 + \sum_{i=0}^{t-1} \eta_i^2 \|\boldsymbol{g}_i\|^2 + 2\sum_{i=0}^{t-1} \eta_i \langle \bar{\boldsymbol{g}}(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle + 2\sum_{i=0}^{t-1} \eta_i h_i(Z_i),$$

where

$$\boldsymbol{g}_i := \boldsymbol{g}(\boldsymbol{\theta}_i, Z_i), \qquad h_i(z) := \langle \boldsymbol{g}(\boldsymbol{\theta}_i, z) - \bar{\boldsymbol{g}}(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle .$$

The mean drift is helpful:

$$\langle \bar{\boldsymbol{g}}(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle = -\big[f(\boldsymbol{\theta}_i) - f(\boldsymbol{\theta}^*)\big] \le 0.$$

Under IH_{t-1}, the quadratic term is controlled because $\sum_i \eta_i^2 < \infty$. The only hard term is the Markov bias

$$B_t := \sum_{i=0}^{t-1} \eta_i h_i(Z_i).$$

Stability proof II: use the Poisson equation for Markov bias

Freeze $\boldsymbol{\theta}_i$. Since $\mathbb{E}_{Z\sim\pi_Z}[h_i(Z)] = 0$, solve the Poisson equation

$$u_i - P_Z u_i = h_i, \qquad u_i(z) = \sum_{\ell=0}^{\infty} (P_Z^{\ell} h_i)(z), \qquad \|u_i\|_\infty \le 16\tau_{\mathrm{mix}} \|h_i\|_\infty .$$

Then each Markov-noise term decomposes as

$$h_i(Z_i) = \underbrace{u_i(Z_{i+1}) - (P_Z u_i)(Z_i)}_{\text{martingale difference}} + \underbrace{u_i(Z_i) - u_i(Z_{i+1})}_{\text{telescoping remainder}} .$$

Under IH_{t-1},

$$\|h_i\|_\infty \lesssim \phi_\infty^2 R_{\max}^2, \qquad \|u_i\|_\infty \lesssim \tau_{\mathrm{mix}} \phi_\infty^2 R_{\max}^2.$$

Two key implications

- The first term is controlled by Pinelis' inequality for bounded differences.
- The second term is rewritten by Abel summation; the time-varying u_i's are controlled by Lipschitz continuity and $\boldsymbol{\theta}_i - \boldsymbol{\theta}_{i-1} = \eta_{i-1}\boldsymbol{g}_{i-1}$.

Stability proof III: close the bootstrap

The previous bounds give

$$\|\boldsymbol{\theta}_t\|^2 \le \left(4 + \frac{A_1(\delta)\rho^2}{c} + \frac{A_2\rho^2}{c^2}\right) R_{\mathrm{base}}^2.$$

To close the induction, it is enough that

$$4 + \frac{A_1(\delta)\rho^2}{c} + \frac{A_2\rho^2}{c^2} \le \rho^2.$$

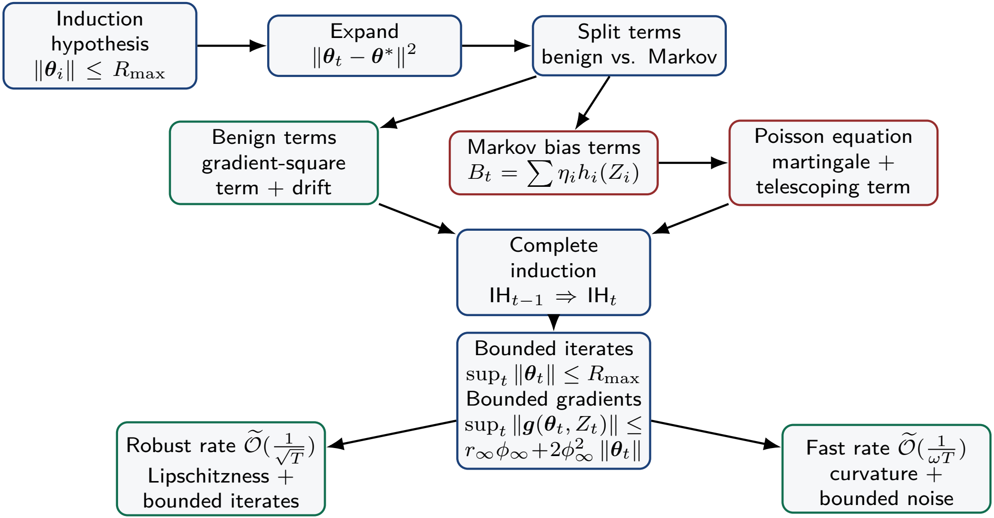
Many pairs (c, ρ) close the induction

- As $c \to \infty$, the admissible radius factor satisfies $\rho \downarrow 2$.
- There is a minimum requirement $c > c_{\min}(\delta)$.
- Larger c: smaller stepsize, smaller radius bound.
- Smaller c above the threshold: larger stepsize, larger radius bound.

Bootstrap conclusion

$$\mathsf{IH}_{t-1} \Longrightarrow \mathsf{IH}_t, \qquad \sup_{t\ge 0} \|\boldsymbol{\theta}_t\| \le \rho R_{\mathrm{base}}.$$

Proof roadmap: stability first, rates second



Conclusion

Main message

A single ω-agnostic stepsize suffices for projection-free linear TD(0) under Markov sampling.

- **Algorithmic takeaway:** no projection radius and no tuning using the curvature/eigenvalue parameter ω.
- **Statistical takeaway:** the averaged iterate achieves simultaneous high-probability rates
$$\min\left\{\widetilde{\mathcal{O}}\left(\frac{1}{\omega T}\right),\ \widetilde{\mathcal{O}}\left(\frac{1}{\sqrt{T}}\right)\right\}.$$
- **Proof takeaway:** self-bounding induction gives implicit boundedness; a Poisson-equation decomposition controls Markov bias.

Open problem: Can we still get a reasonable best-of-both-worlds rate with a τ_{mix}-agnostic stepsize?

- The current stepsize is $\eta_t = \frac{1}{c\,\tau_{\mathrm{mix}}\,\phi_\infty^2}\,\frac{1}{\sqrt{t+1}\log(t+3)}$, but τ_{mix} is usually unknown in practice.
- A mixing-free stepsize is available: $\eta_t = \frac{1}{c\,\phi_\infty^2}\,\frac{1}{\sqrt{t+1}\log^2(t+3)}$, but it gives a doubly exponential dependence on τ_{mix} in the constants.

Thank you! Any questions?

Slides

Paper

Personal website

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